

Overview of Computational methods for Bayesian inference

Joshua French

Introduction

Summary of computational methods

- Bayesian statistics focuses on determining or summarizing the posterior distribution.
- When a closed-form expression for the posterior distribution is unavailable, we rely on numerical approximations.

Deterministic approximations

- Maximum a posteriori estimation
- Cubature-type integration approaches
- Distributional approximations

Stochastic approximations

- Direct (Monte Carlo) simulation
- Markov chain Monte Carlo (MCMC) simulation

Maximum a posteriori estimation

- The *maximum a posteriori* estimator summarizes the posterior with the parameter value(s) that maximize the posterior distribution.
 - $\hat{\theta}_{\text{MAP}} = \operatorname{argmax}_{\theta} \ln p(\theta|y) = \operatorname{argmax}_{\theta} \{\ln p(y|\theta) + \ln p(\theta)\}.$
- In practice, we can find this numerically using optimization functions built-in to our software of choice.

Cubature integration

- The quantities of interest can often be written as an integral.
 - We can use fast deterministic methods to evaluating the integral.
 - E.g., quadrature rule, Simpson's rule, etc.

- The posterior expectation of any function $h(\theta)$ is defined as

$$E[h(\theta) | y] = \int h(\theta)p(\theta | y)d\theta.$$

- E.g., $p(\tilde{y} | y) = E_{\theta|y} [p(\tilde{y} | \theta, y)] .$

Distributional approximations

- The posterior distribution can be approximated by another known distribution.
- The most common choice approximates the posterior distribution by $\theta | y \sim N(\hat{\theta}, I(\hat{\theta})^{-1})$.
 - a.k.a., the Bayesian Central Limit Theorem
 - $\hat{\theta}$ is an estimate of θ such as the MLE or MAP.
 - $I(\hat{\theta})$ is the *observed information matrix*.

Observed information matrix

- The observed information matrix is

$$I(\hat{\theta}) = -\mathbf{H}(\theta) \mid \theta = \hat{\theta},$$

where

$$\mathbf{H}_{i,j} = \frac{\partial^2}{\partial \theta_i \partial \theta_j} \ln p(y \mid \theta).$$

- $\hat{\theta}$ is chosen to be the MLE of θ from the data or the MAP estimate from the posterior.

Stochastic methods

- Stochastic methods for approximating the posterior rely on the law of large numbers (LLN).
- The LLN says that if $\theta^{(1)}, \dots, \theta^{(B)}$ are i.i.d. realizations from $p(\theta \mid y)$, then for (almost) any function $h(\theta)$, as $B \rightarrow \infty$:

$$\frac{1}{B} \sum_{j=1}^B h(\theta^{(j)}) \rightarrow E[h(\theta \mid y)] = \int h(\theta) p(\theta \mid y) d\theta.$$

- The LLN (generally) applies to simulated data that has a Markov structure, i.e., each simulated value only relies on knowing the previous simulated value.

Other Distributional Approximations

- Integrated nested Laplace approximation (INLA)
 - A more advanced normal-based approximation to the posterior that has gained popularity.
- Variational Bayes methods
 - The methods postulate a possibly non-Gaussian parametric distribution for the posterior and then solve for the parameter estimates that make the distributions most similar.

LLN implications

- $\bar{\theta} = \frac{\sum \theta^{(j)}}{B} \rightarrow E(\theta \mid y)$
- $\frac{1}{B-1} \sum (\theta^{(j)} - \bar{\theta})^2 \rightarrow \text{var}(\theta \mid y)$
- $\frac{1}{B} \sum_{j=1}^B I(\theta^{(j)} \leq c) \rightarrow P(\theta \leq c \mid y)$
- The empirical distribution of $\{\theta^{(1)}, \dots, \theta^{(B)}\} \rightarrow p(\theta \mid y)$
- The α -quantile of $\{\theta^{(1)}, \dots, \theta^{(B)}\}$ converges to the α -quantile of $p(\theta \mid y)$

Direct (Monte Carlo) simulation

- Direct simulation approximates the posterior by drawing i.i.d. samples from the posterior.
- In simple contexts, this can be done using *rejection sampling* (also known as *acceptance-rejection sampling*).
 - This tends to not scale very well to high-dimensional contexts.

Direct simulation (cont)

- *Approximate Bayesian computation* (ABC) attempts to perform rejection sampling without evaluating the likelihood function.
 - In principle, this will be MUCH faster.
 - Possibly erroneous assumptions must be made.

MCMC simulation

- MCMC methods produce correlated samples from the posterior.
- The Gibbs sampler is often the preferred MCMC method.
 - Samples are drawn from the full conditional distributions of each parameter.
 - E.g., Samples are drawn for $p(\theta_1, \theta_2 \mid y)$ by drawing a sample from $p(\theta_1 \mid y, \theta_2)$, then from $p(\theta_2 \mid y, \theta_1)$, and repeating this process.

MCMC simulation (cont)

- The Metropolis-Hastings algorithm is a more general MCMC method.
 - A sample is drawn from a proposal distribution for the parameter.
 - The sample is retained with some probability determined by an acceptance ratio.
 - This process is repeated iteratively for all parameters.
- More advanced Monte Carlo approaches, such as Hamiltonian Monte Carlo, are more efficient, but are also more complex.

Probabilistic Programming Languages

- In practice, people use flexible, efficient probabilistic programming languages (PPLs) to draw samples from the (approximate) posterior.
- Popular choices include:
 - Stan (compatible with R, Python, etc.)
 - Nimble (R specific)
 - PyMC (Python specific)

Conclusions

- Bayesian inference requires us to characterize the posterior distribution and quantities derived from it.
- **Computational methods make Bayesian inference possible** when analytical solutions are unavailable.
- Computational methods span deterministic approximations to simulation-based approaches.
- There is no best computational method.
- PPLs make complex Bayesian analysis accessible without requiring us to implement the underlying algorithms from scratch.